Calculus I. Partial 1: Limits and the derivative as a limit process

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Class conditions

- There are no dumb questions
- Cellphones at the professor's desk at the begginig of class
- If you want to use the cellphone, take it from the desk and go outside
- Bring notebook
- All activities and homeworks are deliver by hand
- Late deliveries: Two days after due date with grade over 80
- No electronic devices
- Web assign is for home
- Most of homeworks are webassign and exam corrections

Introduction to the course

Important rules You need to leave your cellphones at the professor's desk at the beginning of the session. If you leave your cellphone at the professor's desk at the beginning of the session, you will get the attendance. If you really need to use it, take it and go outside the classroom. If you use your cellphone during the first part of the session for more than 4 minutes, you will not receive the attendance. If you are here for the first part of the sessions and then you leave after the break, you still have attendance. However, if it is a phone call from your parents, that might be an exception. I have mandalas if you need to do something with your hands.

Main time distribution Generally the classes will be divided into three sections: lecture, break, and workshop. During the lecture time block, I'll introduce the topics, show procedures, or provide feedback, depending on the day. The break will be 5 minutes, and you can use your cellphone, go to the restroom, or take a quick snack. Finally, in the workshop session, there are going to be 3 activities with due dates before ending class. It doesn't matter if you arrive at the correct answer; the most important thing is that you practice math. The lecture and workshop sessions are planned to be 20 minutes each but might change depending on the topics; meanwhile, the break will be 5 minutes.

Exams will be on Monday. Fridays will be more like a recap session and workshop to resolve questions related to WebAssign activities. Now, there is an announcement in Canvas about how to enroll yourselves into the WebAssign platform; you will have until this Friday, January 16th.

Evaluation system The final grade is composed as follows,

1. Partial 1: 28%
2. Partial 2: 28%

3. Final period exam: 30%
4. Final Period quiz: 4%
5. Final webassign activities: 3%
6. Final period activities: 3%
7. Final period core activity: 4%

Each period will be graded as afollows,

- Partial exam: 50%
- Quizzes: 15%
- Collaborative activities: 15%
- Activities: 10%
- Webassign: 10%

Important dates

- Interpartial exam 1: January 26
- Interpartial exam 2: February 9
- Partial exam 1: February 16
- Interpartial exam 3: March 9
- Partial exam 2: March 23
- Interpartial exam 4: April 27
- Partial exam: May 12

Introductio to Calculus

At some point in the history of humankind there were two open questions: How do you determine the slope of a curve at a single, exact point? And how do you measure the area of a shape with a curved boundary? The first question was asked by the Greeks. They were studying lines that only touch a circle once, most commonly known as the tangent line to a circle. However, for other kinds of graphs it was a mystery, like for a straight line, a parabola, or an ellipse. After some centuries, the answer to this question led us to the development of the derivative. On the other hand, the second question led to the creation of the integral. In a more specific tale, Archimedes was a master of computing areas under parabolas; however, his method was a headache. Some years later, astronomers like Kepler needed to calculate areas of planetary orbits, and this was solved by the definition of an integral.

In general terms, the derivative allows us to give a quantitative answer to questions about the rate of change at an instant, like “How fast is it changing right now”? The integral allows us to give a quantitative answer to questions about the accumulation of change, like “How much has accumulated in total over time”? These two questions are fundamentally linked. If you have a graph of speed over time, the derivative gives the acceleration at any moment, and the integral gives the total distance traveled.

In this course, we only are going to focus on the first question, "*How do you determine the slope of a curve at a single, exact point?*" The answer of this question is going to lead us to the definition of the derivative and that concept it is usefull at GPS navigation, medicine dosing, video games, econocmis, climate models, and other disciplines.

Hence, some of the applications that we are going to explore at the end of the course are related speed, growth, decay, accumulation or in a more abstract concept, flux.

The big picture Calculus is related with arithmetic in the sense that the addition operation finds the total and the substraction operation finds a difference. In calculus, the integration operation finds a total, meanwhile the derivative finds a rate of change. The reason of why calculus is more related with arithmetic is because arithmetic give us tools to quantify **static** amounts, meanwhile calculus give us tools to quantify **dynamic**, changing amounts.

This is quite different from the math that you learn from previous courses. In math 3 and math 4 it was more focused on algebra. And algebra is more like an architect who designs the blueprint, that is the function. Calculus is the surveyor and engineer who uses specialized tool to take precises measurements from that blue print.

For example, algebra give us a function that models the height $h(t) = -5t^2 + 20t + 1$. And can also allow us to answer the question "*When does the ball hit the ground?*" by solving $h(t) = 0$. Calculus is going to help us to answer questions like "*What is the ball's velocity at the exact instant?*" with the derivative and "*What is the total distance traveled by the ball between and interval of time?*" with integration.

As a wrap up, arithmetic allow us to handle numbers. Algebra is to handle unknowns and relationships. And now, calculus will help us to handle change itself. Finally, the main association that I want you to keep in mind is the following: Derivatives answers "*How fast right now*" and the integrals answers *How much in total*. On the other hand, I also want to make an spoiler. The integral and derivative are two sides of the same coin or also known as inverse operations.